Amrit Rajneesh Bahl

amritbahl.github.io amritrbahl@gmail.com LinkedIn

Education

Stanford University Incoming, Fall 2026
Master of Science in Mechanics and Computation — **GPA:** Stanford, CA, USA
• **Relevant Coursework**:

Birla Institute of Technology & Science, Pilani (BITS Pilani) Aug 2021 - May 2025
Bachelor of Engineering in Civil Engineering — **GPA: 9.26/10** Pilani, India
• **Relevant Coursework**: Mechanics of Solids, Analysis of Structures, Design of Reinforced Concrete Structures, Foundation Engineering, Design of Steel Structures, Earthquake Resistant Design and Construction, Introduction to FEM

Technical Skills

Software: DIANA FEA, ABAQUS, ETABS, RocScience (RS2, RS3), Optum (G2 and G3), AutoCAD, Revit, Primavera P6, EPANET, OriginPro, MS Office Suite
Programming Languages: MATLAB, Python, C, LaTeX

Research Experience

Numerical study on RC base-restrained walls and tanks subjected to shrinkage effects Jun 2024 - Present
***Undergraduate Thesis**: Dr. Mohammad Najeeb Shariff, Indian Institute of Technology Bombay*
- Compared existing analytical models in direct tension and restrained shrinkage along with a Sobol's sensitivity analysis to determine the accuracy and effect of the various parameters on crack-width. Submitted the results as a paper to the *4th R.N. Raikar Memorial International Conference 2024*
- Validated the crack-widths, crack-patterns, and shrinkage strains from the experimental tests done on base-restrained walls and tanks using a 3D nonlinear finite element (FE) analysis on DIANA, an FE software
- Fitted an equation to find the variation of the restraint factor along the height of a wall for different aspect ratios derived using a linear FE analysis to assess the tensile stresses due to a base-restraint. The results have been submitted as a paper to the *ACI Structural Journal*

Dynamic stability analysis of randomly distributed CNT reinforced composite beam Jan 2024 - May 2024
***Design Project**: Dr. Rajesh Kumar, BITS Pilani*
- Modeled the effective three-phase CNT-Fiber-Composite system using the Eschelby-Mori-Tanaka scheme and the Chamis homogenization technique
- Solved the governing differential equations assuming the First-Order Shear Deformation theory using the Generalized Differential Quadrature method to validate the non-dimensional natural frequency and critical buckling load
- Derived the dynamic instability regions using Bolotin's method and submitted the results of the parametric study as a book chapter in the book *Experimental and Mathematical Modeling of Aerospace and Civil Engineering Composite Structures*

Susceptibility of soil-rock mixture slopes to Landslides Aug 2023 - Dec 2023
***Design Project**: Dr. Nishant Roy, BITS Pilani*
- Reviewed the existing literature to conduct a bibliometric analysis on modeling slope stability of soil-rock mixtures
- Developed a MATLAB code to generate randomly distributed circular and spherical blocks in an embankment configuration following a gradation of rock particles
- Exported the complicated geometries generated in MATLAB to Optum G2 and Optum G3 through the software's backend to create 2500 models with varying parameters

Stabilization of Black-Cotton Soil with Alkali Bypass dust Jun 2023 - July 2023
***Summer Internship**: Sr. Technical Officer V. K. Kanaujia, CSIR-Central Road Research Institute (CRRI)*
- Performed laboratory tests to determine the index properties of the Black Cotton soil and determine the improvement in properties when mixed with Alkali Bypass Dust based on the strength and plasticity characteristics
- Presented the results of the parametric analysis to determine the optimum mix as a poster in the Young Researchers Conclave at CRRI

Assessing the suitability of brick-dust aggregate to prepare a Geopolymer Jan 2023 - Apr 2023
***Laboratory Project**: Prof. Dipendu Bhunia, BITS Pilani*
- Performed laboratory tests like X-ray fluorescence (XRF), X-ray diffraction (XRD), and Thermo-Gravimetric Analysis (TGA) to chemically characterize the brick-dust fines
- Determined that the brick-dust would be suitable as a precursor to making Geopolymer based on the amount of Si^{4+} and Al^{3+}

CURRICULAR PROJECTS

Design and Analysis of a Multi-Storey Steel Shopping Mall Jan 2024 - May 2024

***Course Project**: Design of Steel Structures*

- Designed and analyzed a G+3 steel shopping mall structure in ETABS using the Limit State method as per IS 800:2007
- Checked the design of critical members using hand calculations as per IS 800:2007

Design of a standalone apartment Jan 2023 - May 2023

***Course Project**: Construction Planning and Technology*

- Drafted the plan of a G+0 residential apartment on AutoCAD and modeled the 3D structure with plumbing on Revit based on the National Building Code of India 2016
- Conducted a cost and time analysis of the apartment construction to determine the required resources using Primavera P6 and prepared a schedule for the construction activities

TEACHING ASSISTANTSHIPS

Soil Mechanics Jan 2024 - May 2024

***CE F243**: Dr. Sayantan Chakraborty*

- Explained the 1D consolidation test, through weekly demonstrations, to a group of 50 undergraduate students
- Responsible for the invigilation of evaluative components and the doubt-clearing sessions of around 90 undergraduate students enrolled in the course

Fluid Mechanics Aug 2023 - Dec 2023

***CE F231**: Dr. Selva Balaji*

- Prepared questions and their solutions on each of the 10 modules covered in the course
- Responsible for the invigilation of evaluative components and the doubt-clearing sessions of around 90 undergraduate students enrolled in the course

PUBLICATIONS

- *Experimental and numerical studies on reinforced concrete base restrained walls and tanks subjected to shrinkage effects* Amrit Bahl, Mohammad Najeeb Shariff, and Sankati Yellamanda
  Paper submitted to the *ACI Structural Journal*
- *A comparison of analytical models for crack-width prediction in RC ties subject to direct tension and restrained shrinkage* A R Bahl, Pawan Prakash, and M N Shariff
  Accepted in the *4th R.N. Raikar Memorial International Conference 2024*
- *Dynamic stability of randomly distributed CNT reinforced composite beam using the generalized differential quadrature method* A R Bahl, and Rajesh Kumar
  Submitted as a book chapter in the book *Experimental and Mathematical Modeling of Aerospace and Civil Engineering Composite Structures by Apple Academic Press (CRC Press/ Taylor and Francis)*

LEADERSHIP AND EXTRA CURRICULARS

Student Alumni Relations Head | *Head of Events*

- Led a team of 25 members to organize the batch photograph and the Director's Tea Party for the graduating batch of around 900 students
- Coordinated and organized the reunion of 3 alumni batches hosting around 100 alumni from each batch

Hindi Press Club | *Senior Editor*

- Served as a writer and then the editor of the monthly magazines and the special issues published by the Hindi Press Club of BITS Pilani circulated among 4000+ students and faculty members
- Worked in a team of 30 members to ensure the coverage of all events during the cultural, technical, and sports festival at BITS by publishing daily newsletters read by students and participants

Student-Faculty Committee | *Student Representative*

- Selected as one of the 5 representatives from a batch of 90 students to address the academic and non-academic issues of the student body and find effective solutions through bi-semesterly meetings
- Presented the students' standpoint on potential electives, workshops, and activities conducted in the department

National Service Scheme | *Volunteer*

- Conducted bi-weekly lectures on English and Personality Development for underprivileged children from classes 6^{th} to 12^{th}
- Organized the Old Notebook Collection Drive with 5 team members by coordinating with 100+ volunteers and collected 200 kilograms of scrap paper

ACADEMIC ACHIEVEMENTS

Poster Presentation Competition - Young Researchers Conclave

- Awarded the second position in the poster presentation competition organized as a part of the Young Researchers Conclave at the CSIR-CRRI for the poster titled *Stabilization of Black Cotton Soil with Alkali Bypass dust*